

JACOB COOK

Security Engineer III | IR & Vulnerability Mgmt | SIEM · Pen Testing · Cloud · IDS/IPS · Network & Endpoint Hardening

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Hands-on security engineer with breadth across detection, IR, vulnerability management, and cloud/identity hardening. Built a Wazuh SIEM from scratch, authored Sigma rules mapped to MITRE ATT&CK, and engineered Python/FastAPI automation for IR. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX) with pen-testing fundamentals, and an MS Cybersecurity in progress. BS Computer Science. Strong communicator able to translate technical security issues for non-technical audiences.

TECHNICAL SKILLS

Detection, IR & SIEM: Wazuh SIEM (build & admin), Sigma Rule Authoring, MITRE ATT&CK, Threat Intelligence Concepts, IOC Analysis, IR Lifecycle, Microsoft Sentinel (familiar), KQL/SPL (familiar), Endpoint Protection Concepts, IDS/IPS Concepts
Vuln Mgmt & Pen Testing: CompTIA PenTest+ certified, Vulnerability Assessment Concepts, Remediation Coordination, OWASP, Security Hardening, Network Security (TCP/IP, DNS, SSL/TLS)
Cloud & Identity: Azure AD / Entra ID, Okta, AWS labs (IAM, encryption, monitoring), SSO / RBAC, MFA, Terraform, DevSecOps Practices, Zero Trust Concepts
Automation & Scripting: Python, Bash, SQL, FastAPI, Webhook-Driven Playbooks, Git

TECHNICAL PROJECTS

- Stryker / Intune Detection Pack: Detection-as-Code** 2026
- Designed and implemented 6 Sigma-format detection rules mapped to MITRE ATT&CK that eliminated 100% of false positives in a 100-event simulation — direct monitoring/detection mechanism design work.
 - Documented 18 benign false-positive scenarios as repeatable evidence-trail artifacts.
 - Verification: 43 passing pytest cases in repo (tests/test_rules.py).
- Blue Team SOC Monitoring Lab: Wazuh SIEM** 2026
- Built a full Wazuh SIEM stack from scratch and validated 12 detection patterns against 242 simulated events with 100% precision after tuning — direct SIEM configuration/maintenance and log analysis work.
 - Reduced false-positive volume by 100% by tuning out 79 benign-only matches with documented rationale.
 - Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).
- SOAR-lite: Incident Response Orchestrator (Python / FastAPI)** 2026
- Automated incident response workflows: webhook ingestion, playbook routing, IP blocking, Slack notification, and incident write-up — direct security orchestration and automation work.
 - Eliminated four manual triage steps and reduced triage time by over 60% in lab testing.
 - Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py).
- Phishing Analysis Lab** 2026
- Triaged 5 phishing samples end-to-end with header inspection, SPF/DKIM/DMARC validation, IOC extraction, and attacker infrastructure mapping — 100% accurate verdicts.
 - Translated triage into IOC-driven detection improvements feeding back into SIEM rule tuning.
- G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)** 2024
- Architected RBAC and Azure AD SSO controls on a production AI platform — direct cloud security and IAM implementation in a regulated stakeholder environment.
 - Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

- Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response** 2019 - 2025
- Maintained 99.5% uptime by triaging telemetry on rotating shifts — six years applying continuous-monitoring and detection discipline expected of a senior security engineer.
 - Resolved 200+ anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30% — direct cross-functional communication with engineering and operations.
 - Reduced mean time to recovery by an estimated 20% via cross-team coordination, escalation, and post-incident reporting.

EDUCATION & CERTIFICATIONS

- WGU | MS, Cybersecurity (In Progress)** Expected Oct 2026 · GPA 3.50
- CSUS | BS, Computer Science** Jan 2025 · GPA 3.50
- CompTIA:** Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)